
Where Does Latent Planning Fail?

Controlled Bottleneck Attribution for World-Model Planning

Anonymous Author(s)

Affiliation

Address

email

Abstract

When a latent world-model planner selects a poor action, inaccurate prediction is a natural suspect, but decisions also depend on how predicted futures are scored and which candidates search exposes. We introduce a controlled attribution protocol that changes one interface at a time and executes every selected action in the simulator. Candidate-matched oracle substitutions estimate recoverable prediction, decision-metric, and search headroom using initial states as the independent units. A controlled local Push-T protocol exhibits negligible prediction headroom but a substantial metric gap; the diagnosis-selected scorer reduces normalized regret by 0.0815 (95% CI [0.0563, 0.1079]). A disjoint-state prospective Wall validation reverses the diagnosis: prediction headroom is 0.5636 [0.4871, 0.6373], and a pre-specified predictor repair improves regret by 0.3891 [0.2930, 0.4804], while a wrong-layer metric repair does not. On the official high-dimensional Push-T H6 CEM trace, both gaps are positive and the prediction-gap point estimate is larger under a frozen classification rule. A scorer stress test is directionally positive but statistically inconclusive and changes adaptive search coverage. Across these protocols, the ordering of recoverable headroom changes with the planning regime. These results support a practical rule: diagnose where decision quality is lost before retraining the world model or changing the planner.

1 Introduction

Latent world models make planning computationally attractive by predicting future representations instead of reconstructing pixels [Hafner et al., 2019, Zhou et al., 2025, Assran et al., 2025, Terver et al., 2026]. A planner, however, contains more than a predictor. Candidate actions are proposed, future latents are predicted, a decision metric ranks them, and search selects an action for execution. A poor action can therefore arise even when predicted latents are close under the training loss. Conversely, visibly imperfect predictions can preserve enough information to select the best available action.

This distinction matters in practice. Predictor retraining may require new data and substantial compute. If recoverable decision quality is instead lost at the scoring or proposal interface, predictor improvements target the wrong component. The reverse error is equally possible: a task-aligned scorer cannot restore information that prediction has discarded. Generic prediction accuracy alone is therefore insufficient to localize a planning failure.

We ask: *where is the dominant recoverable bottleneck in a frozen world-model planning pipeline?* PhysPlan answers with controlled oracle substitutions. For the same initial state and candidate actions, it replaces predicted futures with encoded simulator futures, replaces latent distance with a frozen task readout, or compares queried actions with simulator oracles. Every resulting selection is evaluated

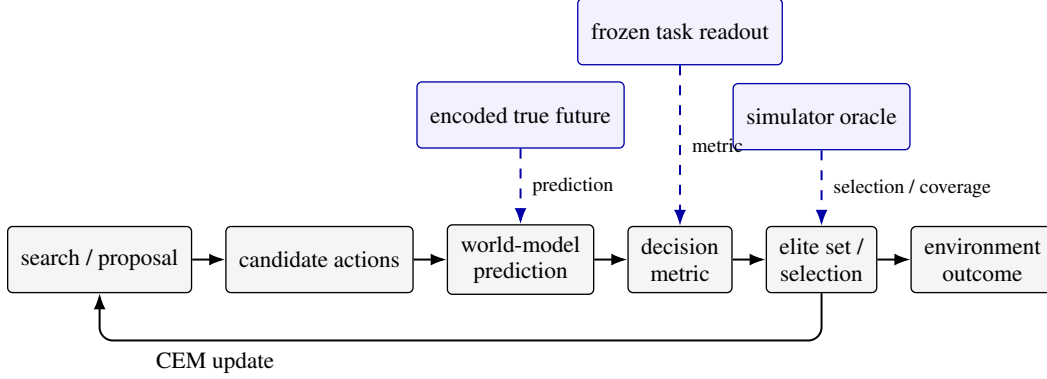


Figure 1: PhysPlan attributes recoverable decision headroom by controlled substitutions at planner interfaces. On a fixed candidate trace, encoded true futures isolate prediction headroom and a frozen task readout isolates metric headroom. Simulator oracles expose within-search selection and proposal coverage. Under adaptive CEM, elite selection updates the proposal, so a scorer intervention changes both ranking and future candidate coverage.

by simulator cost. These contrasts estimate the decision headroom recoverable at prediction, metric, proposal, and within-search selection interfaces (Figure 1); they are diagnostic contrasts, not an additive causal decomposition.

We test whether a diagnosis prospectively identifies a useful intervention. In a controlled local Push-T protocol, prediction headroom is nearly zero while decision-metric headroom is substantial; a pre-specified metric repair recovers a similar amount of regret, whereas increasing predictor inference precision does not recover planning performance. A disjoint-state prospective Wall validation yields the opposite diagnosis, and its pre-specified predictor repair succeeds on unseen states while a wrong-layer metric repair does not. Finally, the official high-dimensional Push-T H6 planner reveals a different ordering: both fixed-trace gaps are positive, the prediction-gap point estimate is larger, and the frozen rule classifies prediction as important. The scorer also changes the candidate distribution explored by adaptive CEM [Rubinstein, 1999]; this scorer stress test is directionally positive but statistically inconclusive.

Our contributions are threefold:

1. a candidate-matched, simulator-grounded protocol that turns familiar oracle substitutions into a prospective workflow: freeze interface gaps, select a repair, and test it against a wrong-layer control;
2. cross-protocol evidence that controlled local Push-T is metric limited whereas a disjoint-state Wall validation is strongly prediction limited, with evidence of different strength made explicit; and
3. an official Push-T H6 study showing mixed fixed-trace headroom and metric–search coupling: under adaptive CEM, changing the scorer changes both candidate ranking and future search coverage.

We do not claim to invent oracle ablation, task-aware evaluation, a world-model architecture, or a latent objective. The narrower contribution is to use candidate-matched substitutions as a precommitted diagnosis–intervention test across planning regimes, while separating fixed-trace attribution from adaptive-search effects.

2 Related Work

World models for control. Learned dynamics models support planning from pixels or compact state representations, from PlaNet and probabilistic ensemble control to recent scalable latent-control systems [Hafner et al., 2019, Chua et al., 2018, Hansen et al., 2024]. DINO-WM plans over frozen visual features without reconstructing observations [Zhou et al., 2025]; V-JEPA 2 and JEPA-WMs

study action-conditioned joint-embedding prediction for physical planning [Assran et al., 2025, Terver et al., 2026]. These works establish strong model-and-planner pipelines. We instead hold a pipeline fixed and ask which interface offers recoverable decision headroom.

Prediction quality and decision utility. Objective-mismatch studies show that forward prediction error need not track control quality [Lambert et al., 2020], while planning-aware metrics show that prediction errors of similar generic magnitude can have different downstream consequences [Ivanovic and Pavone, 2021]. Value-aware loss functions, value-targeted regression, and value-equivalent models likewise formalize that models should preserve information relevant to decisions rather than optimize generic one-step prediction alone [Farahmand et al., 2017, Ayoub et al., 2020, Grimm et al., 2020, Voelcker et al., 2025]. PhysPlan does not claim this premise or task-aligned scoring as new. It addresses the operational question that precedes method choice: for a frozen planner, which interface has recoverable decision headroom?

Oracle diagnostics and our distinction. Replacing a learned component with ground truth is a standard diagnostic instrument, and an isolated oracle comparison can resemble an ablation. Our contribution is the experimental organization around it: all conditions share candidate actions and simulator evaluation; gaps, thresholds, repair choice, and controls are frozen before repair evaluation; the targeted repair is compared with a wrong-layer control; and the independent unit is the initial state rather than individual candidates. This supports prospective localization under explicit substitutions, not a unique causal decomposition. The Wall validation further separates diagnosis, repair training, and repair evaluation by disjoint state sets.

Sampling-based planning. CEM iteratively samples candidates, selects elites, and updates its proposal [Rubinstein, 1999]; this pattern is common in model-based control [Chua et al., 2018, Hafner et al., 2019]. A planner may fail because its queries never cover a good action or because it misranks actions it did query. Retaining every query and simulator-evaluating those candidates separates coverage from within-trace selection. It also exposes an important limit of fixed-trace attribution: once a scorer changes CEM elites, it changes the future proposal distribution as well as ranking.

3 Controlled Bottleneck Attribution

3.1 Planning pipeline and regret

For initial state s , a proposal mechanism supplies candidate action sequences $a \in A_s$. An encoder maps observations to latents; an action-conditioned predictor produces future latents $\hat{z}_s(a)$; and a decision metric $q(s, a)$ ranks candidates. The simulator cost $J_s(a)$, unavailable to the deployed planner, is used only for evaluation. Lower cost is better.

For method m , let $a_m(s) = \arg \min_{a \in A_s} q_m(s, a)$. On a finite candidate set, we report normalized regret

$$\mathcal{R}_m(s) = \frac{J_s(a_m) - \min_{a \in A_s} J_s(a)}{\max_{a \in A_s} J_s(a) - \min_{a \in A_s} J_s(a) + \epsilon}. \quad (1)$$

All headline uncertainty treats the cloned initial state (or H6 episode) as the independent unit. Multiple nearby candidates from one state are never treated as independent observations.

3.2 Candidate-matched oracle substitutions

The diagnostic compares four conditions while preserving the initial state, candidate actions, and simulator evaluation:

GT readout. A frozen task readout scores encoded simulator futures.

Predicted readout. The same readout scores world-model predictions.

Predicted L2. The baseline latent distance scores the same predictions.

Simulator oracle. Simulator cost selects the best candidate, either from the reference set or from actions queried by search.

112 The readout never observes the action; it maps latent transition information to simulator task cost
 113 and is fit only on development states. Thus it cannot exploit a direct action-to-cost shortcut. It is
 114 a diagnostic substitution when used to measure headroom; using it inside a planner is a separate,
 115 deployable scorer intervention.

116 Prediction and metric headroom are the paired state-level contrasts

$$G_{\text{pred}} = \mathbb{E}_s[\mathcal{R}_{\text{pred-readout}}(s) - \mathcal{R}_{\text{GT-readout}}(s)], \quad (2)$$

$$G_{\text{metric}} = \mathbb{E}_s[\mathcal{R}_{\text{pred-L2}}(s) - \mathcal{R}_{\text{pred-readout}}(s)]. \quad (3)$$

117 Proposal headroom compares the simulator-best queried candidate with the best reference candi-
 118 date. Within-search selection headroom compares the planner’s selected queried candidate with the
 119 simulator-best queried candidate. These are recoverable contrasts under specific substitutions. They
 120 overlap, depend on the candidate set and readout capacity, and must not be summed or interpreted as
 121 a unique causal partition.

122 3.3 Prospective diagnosis and intervention

123 Before repair evaluation, each controlled protocol fixes the substitution definitions, dominance rule,
 124 bootstrap, selected intervention, wrong-layer control, and success criterion. Local Push-T uses a
 125 material mean gap of 0.02 and Wall uses 0.05; in each, a material gap must have a confidence
 126 bound above zero and its mean must be at least $1.5 \times$ the runner-up. H6 was frozen separately:
 127 “prediction important” requires the prediction-gap lower bound to exceed 0.02 and its mean to exceed
 128 $1.5 \times \max(G_{\text{metric}}, 0.02)$. These rules classify evidence; they do not test the paired difference
 129 $G_{\text{pred}} - G_{\text{metric}}$. The prospective prediction is directional: repair at the diagnosed interface should
 130 improve over baseline and outperform a repair at a non-dominant layer. All states, including harmed
 131 cases, remain in the analysis.

132 3.4 Fixed traces versus adaptive search

133 On a frozen candidate set, substituting a score isolates ranking quality. Adaptive CEM is different:

$$q \longrightarrow \text{elite selection} \longrightarrow \text{proposal update} \longrightarrow \text{future candidate distribution}.$$

134 Changing the score therefore changes both ranking and search coverage after the first update. Fixed-
 135 trace headroom and end-to-end adaptive-search gain answer different questions; equality between
 136 them is neither assumed nor required.

137 4 Experimental Setup

138 4.1 Controlled local Push-T

139 Push-T is a contact-rich planar manipulation task [Florence et al., 2022]. The controlled local protocol
 140 uses the frozen action-conditioned JEPA world model and 100 fresh cloned anchor states. Each state
 141 has an identical 81-action angular reference family at horizon H6; finite-grid scalar CEM makes four
 142 rounds of 16 queries plus a final-mean query (65 total). An action-blind StandardScaler+ridge
 143 readout ($\alpha = 100$) consumes terminal pooled visual and proprioceptive latents. Its regularization
 144 is selected from five values by five-fold anchor-grouped development Spearman correlation and it
 145 is fit on 80 development anchors using only encoded true futures and simulator costs. Exact state
 146 restoration, action alignment, temporal indexing, and deterministic replay were validated before
 147 attribution. This restricted local action slice enables auditable candidate matching; it is *not* the official
 148 Push-T planner.

149 The diagnosis, metric repair, and inference-precision control were fixed before repair computation.
 150 The repair replaces latent L2 with the frozen readout under matched queries and search seeds. The
 151 control changes predictor inference from BF16 to FP32 while retaining latent L2; it tests numerical
 152 precision only, not a retrained or architecturally improved predictor. Diagnosis and repair reuse the
 153 same 100 final anchors, so agreement in effect size is an internal prospective consistency check rather
 154 than independent replication.

Table 1: Frozen cross-regime results. Gaps and repair effects are mean normalized-regret contrasts with 95% state/episode-bootstrap CIs. Positive repair values are improvements. H6 is not a validated repair because its interval crosses zero.

Regime	Prediction gap	Metric gap	Diagnosis	Targeted intervention result
Local Push-T	0.0071	0.0851	Decision metric	Metric scorer: +0.0815
	[−0.0051, 0.0198]	[0.0608, 0.1109]		[0.0563, 0.1079]
Wall	0.5636	−0.0279	Prediction	Predictor repair (1 seed): +0.3891
	[0.4871, 0.6373]	[−0.0905, 0.0345]		[0.2930, 0.4804]
Push-T H6	0.1608	0.0950	Mixed; frozen rule: prediction-important	Scorer stress test: +0.0843
	[0.0565, 0.2725]	[0.0026, 0.1978]		[−0.00008, 0.1684]; inconclusive

4.2 Disjoint-state Wall validation

Wall uses the frozen DotWall environment/model interface and a controlled one-dimensional, two-segment waypoint family: 41 reference candidates and the same fixed 11 planner queries for every method. Development (60), diagnosis (80), and repair-validation (80) states have disjoint seeds, cloned states, and layouts. An action-blind StandardScaler+ridge readout ($\alpha = 10$) uses terminal goal-delta, absolute-delta, and squared-delta features from visual and proprioceptive latents; regularization is selected by five-fold anchor-grouped development Spearman correlation using true futures only.

The protocol was frozen before diagnosis, and the repair choice was recorded while the repair cache was absent. The predictor-only repair freezes both encoders and fine-tunes the predictor on 48 training anchors, selecting one checkpoint on 12 development anchors after five epochs. It uses AdamW (learning rate 3×10^{-5} , weight decay 0.01) and visual trajectory MSE plus 0.1 times proprioceptive trajectory MSE. The wrong-layer control applies the frozen readout to unmodified BF16 predictions. Both use the same fixed 11 candidates and are evaluated on 80 entirely new states. One training seed is used; state-bootstrap intervals therefore condition on this fitted checkpoint and do not quantify training-seed variability. This is prospective within-project validation on disjoint states, not replication by an external implementation.

4.3 Official high-dimensional Push-T H6

The frozen H6 validation uses the official one-shot CEM setting: 30 iterations, 300 sequences per iteration, 10 elites, 9,000 model queries, and the returned final mean. Each H6 model action contains five 2-D simulator controls, so CEM searches 60 scalars and the plan executes 30 controls. Thirty fresh near-contact initial states use contiguous precommitted seeds with no filtering, retry, or replacement.

Every sequence queried by baseline latent-L2 CEM is retained and simulator-evaluated. GT-readout, predicted-readout, and predicted-L2 rankings on this identical 9,001-sequence trace per episode isolate fixed-trace prediction and metric headroom. The action-blind H6 readout is a StandardScaler+ridge model ($\alpha = 0.1$), selected by five-fold grouped-development-episode MSE and fit only to true development futures. A matched-budget CEM run changes only the scorer to this readout. This scorer test was precommitted from the local study; it is *not* the prediction repair that the H6 diagnosis would select. Because the scorer changes elite updates, the two adaptive traces are not candidate matched; their union is used only as a finite audit reference, never as a global action oracle.

Statistics. Reported intervals are 95% paired nonparametric bootstrap confidence intervals over independent anchor states or episodes. No candidate-level pseudoreplication is used. Wall repair intervals are conditional on its single fitted predictor. Full protocol and artifact details appear in Appendix A.

5 Results

Table 1 and Figure 2 summarize the frozen results. The ordering of recoverable interface headroom differs across the three planning protocols.

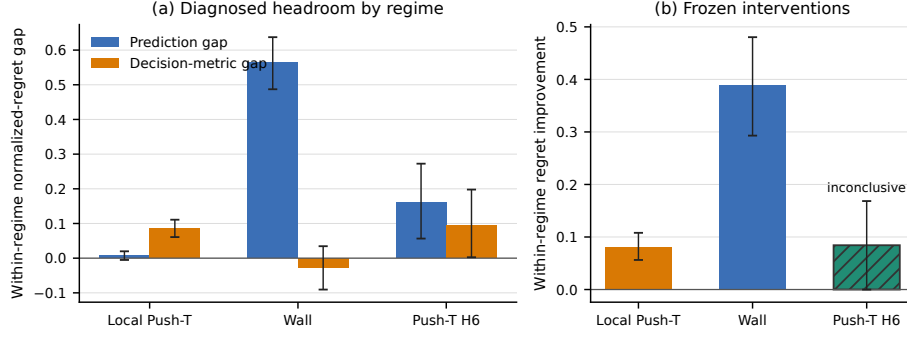


Figure 2: **Across-regime diagnosis and intervention.** (a) Candidate-matched prediction and decision-metric headroom. (b) Mean regret improvement from the local and Wall diagnosis-selected repairs and the H6 scorer stress test. Error bars are 95% paired bootstrap CIs over states or episodes. The hatched H6 result is directionally positive but statistically inconclusive because its CI crosses zero. Regrets are normalized within task-specific candidate sets; compare interface ordering and uncertainty within a regime, not bar magnitude across tasks.

193 5.1 Local Push-T: a decision-metric bottleneck

194 The ground-truth-latent and predicted-latent task readouts achieve state-level Spearman correlations
 195 of 0.696 and 0.689 with simulator cost; their normalized regrets are 0.0289 and 0.0360. Consequently,
 196 prediction headroom is only 0.0071 (95% CI $[-0.0051, 0.0198]$). Under the same predicted futures,
 197 replacing the readout with latent L2 exposes a much larger metric gap of 0.0851 $[0.0608, 0.1109]$.
 198 Proposal headroom is 0.0074. Under the frozen rule, this controlled local protocol is decision-metric
 199 limited.

200 The diagnosis-selected repair reduces regret from 0.1237 with BF16 prediction and latent L2 to 0.0421
 201 with the task readout: an improvement of 0.0815 $[0.0563, 0.1079]$. Across states, 62% improve, 21%
 202 are unchanged, and 17% are harmed. The FP32-predictor plus latent-L2 control improves regret by
 203 only 0.0017, with a CI crossing zero. This supports the narrow statement that increasing predictor
 204 inference precision did not recover planning performance; FP32 is not evidence that every possible
 205 predictor improvement would fail.

206 The observed repair is close to the prior metric gap, whereas the precision control aligns with the
 207 negligible prediction gap. Because diagnosis and repair use the same final anchors and readout, this
 208 numerical closeness is an internal prospective consistency check, not an independent estimate of the
 209 same causal effect. The precommitted repair/control contrast is the principal local result.

210 5.2 Wall: disjoint-state prediction diagnosis

211 On 80 diagnosis states, prediction headroom is 0.5636 $[0.4871, 0.6373]$, metric headroom is -0.0279
 212 $[-0.0905, 0.0345]$, and proposal headroom is 0.0333. The frozen Wall rule therefore diagnoses
 213 prediction as the dominant recoverable interface. This counter-regime shows that the procedure is not
 214 mechanically constructed to declare latent L2 the bottleneck.

215 On 80 entirely new repair states, the pre-specified predictor repair reduces regret from 0.5561
 216 to 0.1670, an improvement of 0.3891 $[0.2930, 0.4804]$. The wrong-layer metric repair improves
 217 regret by 0.0069 $[-0.0489, 0.0625]$. Full-trajectory latent MSE falls from 0.8564 to 0.1999. These
 218 intervals bootstrap evaluation states conditional on one trained repair checkpoint; they do not measure
 219 training-seed variability.

220 Average prediction improvement is not sufficient for state-wise decision improvement. In a retained
 221 counterexample, one anchor’s latent MSE improves while regret worsens from 0.215 to 0.991. This
 222 does not negate the aggregate repair; it bounds the claim: lower predictive error does not guarantee
 223 better action selection for every state. The Wall predictor repair uses one training seed, which limits
 224 claims about training variability.

5.3 H6: changed headroom ordering and metric–search coupling

On the official H6 baseline trace, GT-readout, predicted-readout, and predicted-L2 mean regrets are 0.1724, 0.3332, and 0.4282. Prediction headroom is therefore 0.1608 [0.0565, 0.2725] and metric headroom is 0.0950 [0.0026, 0.1978]. Both gaps are positive; the prediction point estimate is larger and satisfies the frozen prediction-important rule. We did not compute a paired confidence interval for their difference, so this is a rule-based classification rather than evidence that $G_{\text{pred}} > G_{\text{metric}}$ statistically. The ordering differs from the controlled local action slice, but task, candidate dimension, horizon, precision, and planner also change; the study does not isolate which factor causes the shift.

Matched-budget baseline and task-readout CEM obtain regrets of 0.4974 and 0.4131. The scorer-stress-test improvement is 0.0843 [−0.00008, 0.1684]: 16 episodes improve, 9 are unchanged, and 5 worsen. Because the interval narrowly crosses zero, the result is **directionally positive but statistically inconclusive**, not a validated successful repair. Moreover, it tests the local scorer intervention rather than the predictor repair selected by the H6 diagnosis.

The adaptive traces explain why fixed-candidate attribution need not predict end-to-end gain exactly. Changing the scorer reduces within-trace selection gap but can redirect future proposals. In a retained H6 episode, the readout selects the simulator-best sequence in its own trace, yet search coverage degrades and normalized regret worsens from 0.4044 to 1.0000. Thus a better ranking rule on available candidates need not expose better candidates under adaptive search.

6 Discussion and Limitations

What the method adds. Oracle replacement is a familiar ablation device; PhysPlan’s contribution is not the substitution in isolation. The added structure is candidate matching, simulator-grounded regret, frozen classification and repair rules, and targeted-versus-wrong-layer tests. It does not uniquely assign causal responsibility: gaps are practical ceilings under explicit substitutions, and components interact. The value is prospective localization before engineering effort, not an additive causal factorization.

Why not always improve prediction? Prediction is clearly limiting on Wall; in H6, both fixed-trace gaps are positive and the prediction point estimate satisfies the frozen prediction-important rule. It is not the dominant recoverable interface in controlled local Push-T, where a simple metric repair recovers substantial regret and higher predictor inference precision does not. The conclusion is therefore neither “prediction does not matter” nor “metrics are generally limiting,” but that the diagnosis must be repeated when the task, action family, or planner changes.

Scope. Controlled local Push-T uses a restricted local action slice, and Wall uses a controlled 1-D waypoint construction. These designs strengthen candidate-matched attribution but limit behavioral breadth. H6 supplies a realistic high-dimensional planning test, yet contains only 30 episodes; the scorer-test CI narrowly crosses zero, no paired CI compares G_{pred} directly with G_{metric} , and no H6 predictor repair tests the diagnosis-selected interface. Wall’s predictor repair has one training seed, so its state-bootstrap CI is conditional on one learned model. The local Push-T repair shares final states and the readout with diagnosis, although its intervention was committed before computation; only Wall uses a disjoint repair set.

Oracle and interaction limits. The GT-readout condition combines representation observability with the finite capacity and development distribution of the ridge readout; it is a practical ceiling, not a pure representation oracle. Oracle gaps need not be additive. Under CEM, score, elite choice, and proposal coverage interact, so fixed-trace metric headroom is not equivalent to an end-to-end scorer intervention. The search interface is diagnosed but no search-dominant regime and corresponding repair are validated. Cross-task gap magnitudes use task-specific finite-set normalization and candidate families, so they are not a common physical scale.

Generality and reliability. The study covers one latent world-model family and two simulated tasks. “Disjoint-state Wall validation” means prospective evaluation on separate states within this project, not independent reproduction by another group. Differences across protocols descriptively establish changed headroom ordering, but confounded task and planner changes prevent a causal

275 explanation of that migration. The study does not establish broad validity for real robots, vision-
276 language-action models, learned policies, or other planners. Average gains conceal heterogeneous
277 states: both controlled repairs harm some cases, and lower latent MSE does not guarantee state-wise
278 decision improvement. Deployment would require uncertainty-aware gating or fallback mechanisms,
279 which are outside this frozen study.

280 **7 Conclusion**

281 Planning through a latent world model is a multi-interface decision pipeline. Controlled oracle
282 substitutions identify decision-metric headroom in a local Push-T action family and prediction
283 headroom in a disjoint-state Wall validation; pre-specified repairs improve the diagnosed interfaces in
284 those controlled regimes. On the official Push-T H6 CEM trace, both gaps are positive, the prediction
285 point estimate is larger under the frozen rule, and rescoring changes search coverage; the scorer
286 stress test is directionally positive but inconclusive. The supported conclusion is methodological and
287 bounded: recoverable-headroom ordering can change across planning protocols. Diagnose before
288 retraining the world model or changing the planner.

References

- Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning, 2025. URL <https://arxiv.org/abs/2506.09985>.
- Alex Ayoub, Zeyu Jia, Csaba Szepesvari, Mengdi Wang, and Lin Yang. Model-based reinforcement learning with value-targeted regression. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 463–474. PMLR, 2020. URL <https://proceedings.mlr.press/v119/ayoub20a.html>.
- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/3de568f8597b94bda53149c7d7f5958c-Abstract.html.
- Amir-Massoud Farahmand, Andre Barreto, and Daniel Nikovski. Value-aware loss function for model-based reinforcement learning. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1486–1494. PMLR, 2017. URL <https://proceedings.mlr.press/v54/farahmand17a.html>.
- Pete Florence, Corey Lynch, Andy Zeng, Oscar A Ramirez, Ayzaan Wahid, Laura Downs, Adrian Wong, Johnny Lee, Igor Mordatch, and Jonathan Tompson. Implicit behavioral cloning. In *Proceedings of the 5th Conference on Robot Learning*, volume 164 of *Proceedings of Machine Learning Research*, pages 158–168. PMLR, 2022. URL <https://proceedings.mlr.press/v164/florence22a.html>.
- Christopher Grimm, Andre Barreto, Satinder Singh, and David Silver. The value equivalence principle for model-based reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 5541–5552, 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/3bb585ea00014b0e3ebe4c6dd165a358-Abstract.html.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2555–2565. PMLR, 2019. URL <https://proceedings.mlr.press/v97/hafner19a.html>.
- Nick Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations*, pages 47376–47405, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/cf73d57b6dcda32b293df7c2d5341f49-Abstract-Conference.html.
- Boris Ivanovic and Marco Pavone. Injecting planning-awareness into prediction and detection evaluation, 2021. URL <https://arxiv.org/abs/2110.03270>.
- Nathan Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, volume 120 of *Proceedings of Machine Learning Research*, pages 761–770. PMLR, 2020. URL <https://proceedings.mlr.press/v120/lambert20a.html>.
- Reuven Y. Rubinstein. The cross-entropy method for combinatorial and continuous optimization. *Methodology and Computing in Applied Probability*, 1:127–190, 1999. doi: 10.1023/A:1010091220143.
- Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models?, 2026. URL <https://arxiv.org/abs/2512.24497>.

- 339 Claas A Voelcker, Anastasiia Pedan, Arash Ahmadian, Romina Abachi, Igor Gilitschenski, and
340 Amir-Massoud Farahmand. Calibrated value-aware model learning with probabilistic environment
341 models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume
342 267 of *Proceedings of Machine Learning Research*, pages 61745–61768. PMLR, 2025. URL
343 <https://proceedings.mlr.press/v267/voelcker25a.html>.
- 344 Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained
345 visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on*
346 *Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 79115–79135.
347 PMLR, 2025. URL <https://proceedings.mlr.press/v267/zhou25t.html>.

A Detailed Protocol and Reproducibility

A.1 Frozen experimental record

The manuscript is produced from a completed and frozen experimental record; no experiments were rerun for paper preparation. The final code state is commit ceb6122, and the official H6 protocol was frozen at commit 3cbd3c5. Repository validation records confirm 30/30 paired H6 JSON/NPZ traces, 18/18 related tests, 53/53 immutable historical scientific artifacts, and 117/117 SHA-256 manifest checks. Commit hashes are provided without an identifying repository URL to preserve anonymity.

The authoritative machine-readable summaries are:

- controlled local Push-T diagnosis and repair summaries;
- Wall diagnosis and repair summaries, with separate development, diagnosis, and repair-state manifests; and
- the official H6 aggregate summary plus all 30 paired per-episode raw traces.

Each headline value in the paper is mapped to its report section and saved artifact in the anonymized supplementary evidence audit.

A.2 Candidate sets and intervention controls

In controlled local Push-T, every condition uses the same 81-action reference family and the same 65-query budget per anchor. In Wall, all readout comparisons within a diagnosis state use the same one-dimensional waypoint candidates. For fixed-trace H6 attribution, all scores rank the identical 9,001 sequences retained from the baseline trace. Consequently, each fixed-trace regret contrast measures selection on matched candidates rather than proposal quality.

The adaptive H6 comparison is deliberately labeled differently. Baseline and readout CEM share horizon, population, elite count, query budget, initialization, and random-seed policy, but their candidate traces diverge once scores change the elite set. The trace union is a finite diagnostic reference, not an oracle over the continuous action space.

A.3 Readouts and predictor repair

All readouts are action blind and are trained only on encoded true futures from development data. Local Push-T uses terminal pooled visual and proprioceptive latents with StandardScaler+ridge ($\alpha = 100$), selected from $\{0.1, 1, 10, 100, 1000\}$ by five-fold anchor-grouped mean GT Spearman correlation on 80 development anchors. Wall uses terminal goal-delta, absolute-delta, and squared-delta visual/proprioceptive features with ridge $\alpha = 10$, selected from $\{0.1, 1, 10, 100, 1000, 10000\}$ by the same grouped criterion on 60 development anchors. H6 uses terminal pooled visual and proprioceptive latents with ridge $\alpha = 0.1$, selected by five-fold grouped-development-episode MSE. No readout is refit on predicted or final-evaluation latents.

The Wall repair trains only the predictor; both encoders remain frozen. Forty-eight development anchors train the model and 12 select the epoch with minimum validation latent loss. AdamW uses learning rate 3×10^{-5} and weight decay 0.01 for five epochs; the loss is visual-trajectory MSE plus 0.1 proprioceptive-trajectory MSE. The selected epoch is 5. The repair evaluation uses one training seed and 80 disjoint final anchors. Its bootstrap therefore measures evaluation-state uncertainty conditional on that fitted predictor, not variability across training runs.

A.4 Statistical unit and confidence intervals

Local Push-T and Wall bootstrap independent anchor states; H6 bootstraps episodes. Candidate actions and repeated queries within a state are dependent and are not resampled as independent trials. All reported confidence intervals are paired 95% nonparametric bootstrap intervals from the frozen analyses. The repair accounting retains every improved, unchanged, and harmed state or episode.

393 B Additional Frozen Results

Table 2: Supporting frozen measurements omitted from the compact main table. Lower regret and MSE are better.

Regime	Quantity	Value
Local Push-T	GT / predicted readout Spearman ρ	0.696 / 0.689
Local Push-T	GT / predicted readout regret	0.0289 / 0.0360
Local Push-T	FP32 predictor + L2 improvement	0.0017 (CI crosses 0)
Local Push-T	repair outcomes	62% / 21% / 17% improved / same / harmed
Wall	baseline / predictor-repair regret	0.5561 / 0.1670
Wall	wrong-layer metric repair	0.0069 [−0.0489, 0.0625]
Wall	full-trajectory latent MSE	0.8564 → 0.1999
H6	GT-readout / pred-readout / pred-L2 regret	0.1724 / 0.3332 / 0.4282
H6	baseline / scorer-repair regret	0.4974 / 0.4131
H6	repair outcomes	16 / 9 / 5 improved / same / harmed

394 C Claim Boundaries

395 The evidence does not support a universal claim that latent planners are prediction limited or metric
396 limited. It does not make oracle gaps additive or uniquely causal, and the FP32 control does not
397 test a general predictor improvement. The official H6 scorer repair is not statistically significant
398 under the frozen criterion. The controlled local Push-T protocol is not the official planner. Finally,
399 the experiments do not establish transfer to real robots, learned policies, or vision–language–action
400 systems.

401 The evidence also does not establish a statistically significant ordering between the H6 prediction and
402 metric gaps: the prediction point estimate is larger and meets the frozen prediction-important rule, but
403 no paired interval for their difference was computed. The H6 scorer test is not a diagnosis-selected
404 predictor repair. Cross-regime “migration” is descriptive because task, candidate family, horizon,
405 inference precision, and planner differ together.